



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Subject: Digital Signal Processing

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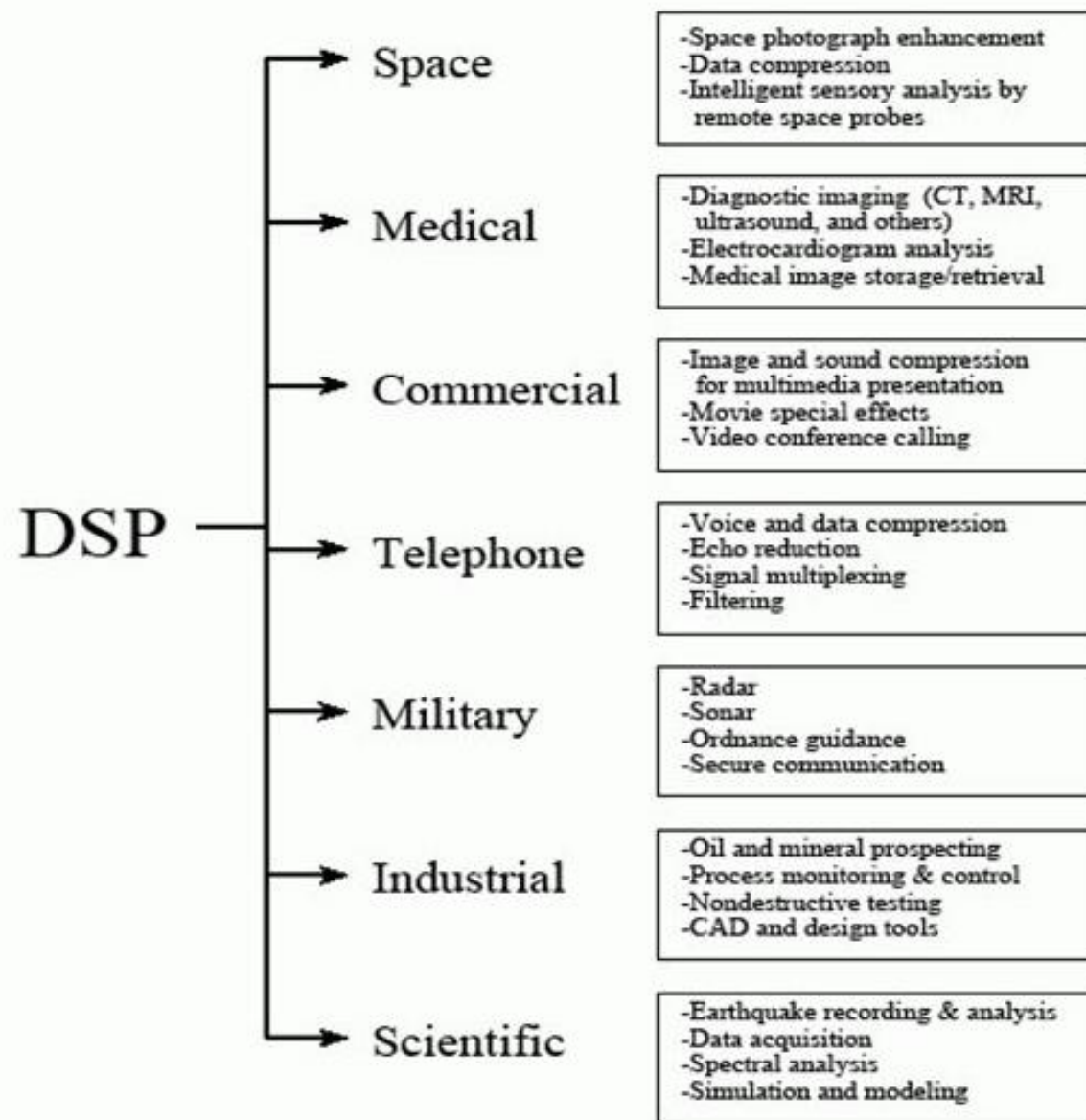
Applications of Digital Signal Processing(DSP)

Some major applications that utilize digital signal processing include:

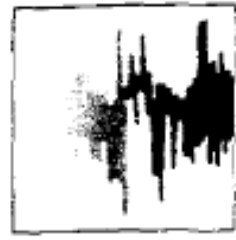
- **Audio processing:** DSP is used in compression of music, equalizers, noise suppressors, echo cancellers, sound spatializes and other audio effects in devices such as mobile phones, music players, smart speakers, headphones, hearing aids, etc.
- **Image processing:** DSP techniques are employed for image enhancement, image restoration, image and video compression, video processing, image segmentation etc. Applications of image processing are digital cameras, medical imaging, satellite images, machine vision, surveillance systems, etc.
- **Speech processing:** DSP is used for speech recognition, voice control, voice search, encoding and decoding of speech, VOIP, speech enhancement and so on and its applications are in mobile phones, smart assistants, handsfree gadgets, hearing aids and so on.

- **Communications:** DSP is essential for modem, cellular, wireless & digital communications in encoding, modulation & demodulation, equalization, error control, multiple access, spread spectrum, synchronization. It is an international term used in radio, mobile networks, broadband among others.
- **Sensors and Control:** DSP techniques can be applied in areas such as sensor fusion and calibration, sensor linearization, motor control, adaptive and stability control, automation / controls, Internet of Things, etc.
- **Signal detection and tracking:** Two important fields of DSP are radar/sonar, and detection/tracking of moving targets, classification of targets, direction of arrival, imaging, moving target indication, noise reduction etc., waveform design, etc..
- **Video processing:** the DSP is implemented for video decoding, interlace to progressive conversion, image stabilization, Video noise reduction, video analytics such as motion detection, object tracking, and recognition in applications such as Home theatre systems and Surveillance cameras.
- **Software Defined Radio:** DSP together with the analog/digital converter technology to make modulation, filtering, and multiple access radio functionalities achievable using software rather than hardware.

APPLICATIONS OF DSP



DSP has revolutionized many areas in science and engineering. A few of these diverse applications are shown here.



2

Discrete-Time Signals and Systems

2.1 DISCRETE-TIME SIGNALS

a discrete-time signal $x(n]$ is a function of an independent variable that is an integer. It is graphically represented as in Fig. 2.1. It is important to note that a discrete-time signal is *not defined* at instants between two successive samples. Also, it is incorrect to think that $x(n]$ is equal to zero if n is not an integer. Simply, the signal $x(n]$ is not defined for noninteger values of n .

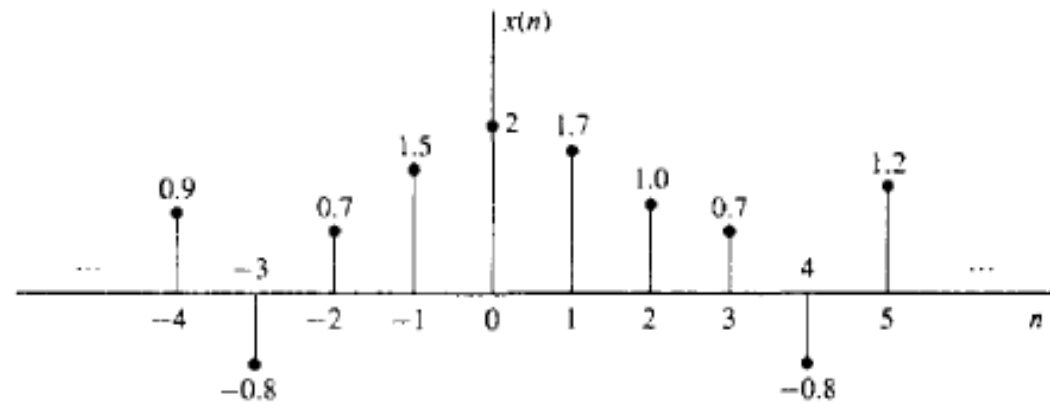
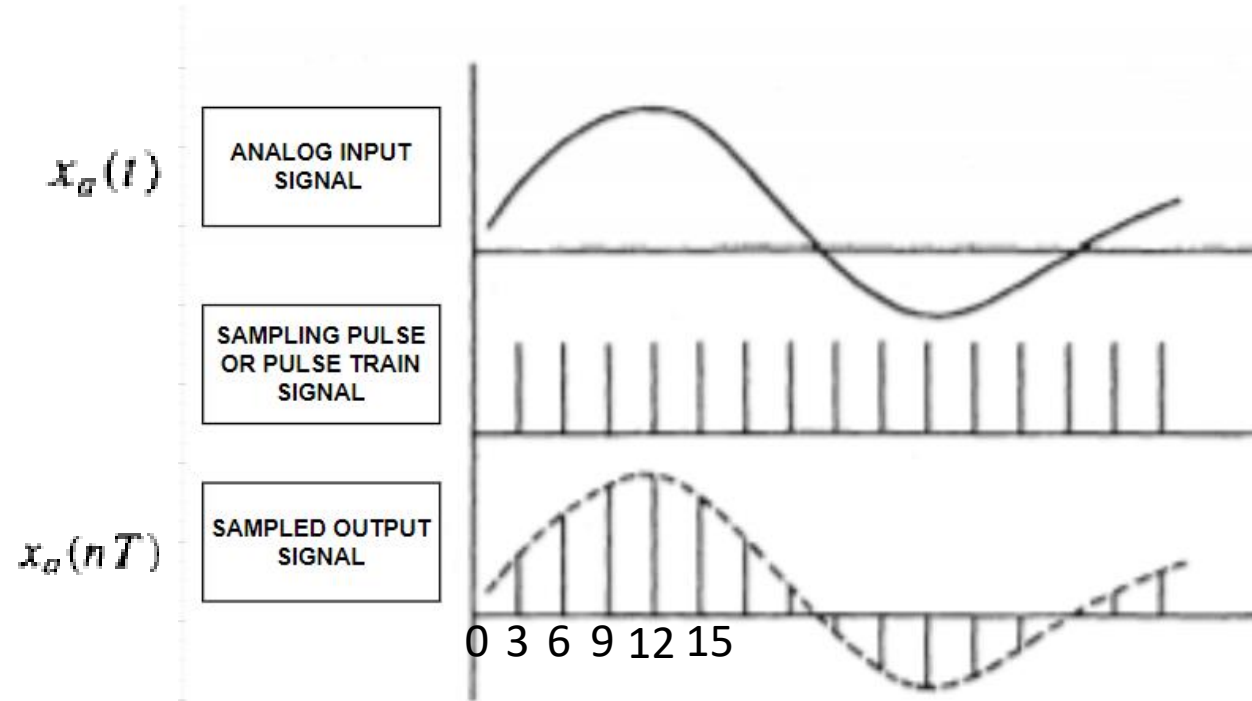


Figure 2.1 Graphical representation of a discrete-time signal.

In the sequel we will assume that a discrete-time signal is defined for every integer value n for $-\infty < n < \infty$. By tradition, we refer to $x(n)$ as the “ n th sample” of the signal even if the signal $x(n)$ is inherently discrete time (i.e., not obtained by sampling an analog signal). If, indeed, $x(n)$ was obtained from sampling an analog signal $x_a(t)$, then $x(n) \equiv x_a(nT)$, where T is the sampling period (i.e., the time between successive samples).



← What will be the value of T ?

Besides the graphical representation of a discrete-time signal or sequence as illustrated in Fig. 2.1, there are some alternative representations that are often more convenient to use. These are:

1. Functional representation, such as

$$x(n) = \begin{cases} 1, & \text{for } n = 1, 3 \\ 4, & \text{for } n = 2 \\ 0, & \text{elsewhere} \end{cases}$$

2. Tabular representation, such as

n	...	-2	-1	0	1	2	3	4	5	...
$x(n)$	...	0	0	0	1	4	1	0	0	...

3. Sequence representation

An infinite-duration signal or sequence with the time origin ($n = 0$) indicated by the symbol $\uparrow$ is represented as

$$x(n) = \{\dots 0, 0, 1, 4, 1, 0, 0, \dots\}$$

$\uparrow$

A sequence $x(n)$, which is zero for $n < 0$, can be represented as

$$x(n) = \{0, 1, 4, 1, 0, 0, \dots\}$$

$\uparrow$

The time origin for a sequence $x(n)$, which is zero for $n < 0$, is understood to be the first (leftmost) point in the sequence.

Show its graphical representation

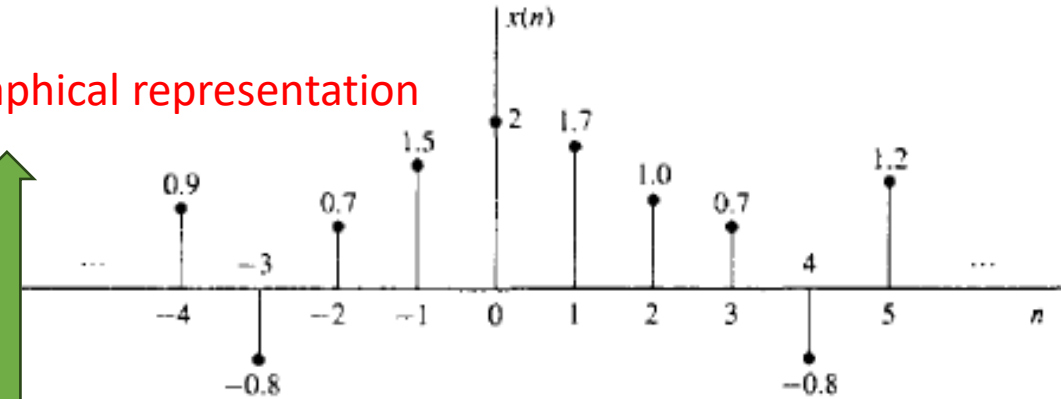


Figure 2.1 Graphical representation of a discrete-time signal.

Show its graphical representation

(2.1.2)

(2.1.3)

A finite-duration sequence can be represented as

$$x(n) = \{3, -1, -2, 5, 0, 4, -1\} \quad (2.1.4)$$

↑

whereas a finite-duration sequence that satisfies the condition $x(n) = 0$ for $n < 0$ can be represented as

$$x(n) = \{0, 1, 4, 1\} \quad (2.1.5)$$

↑

The signal in (2.1.4) consists of seven samples or points (in time), so it is called or identified as a seven-point sequence. Similarly, the sequence given by (2.1.5) is a four-point sequence.

2.1.1 Some Elementary Discrete-Time Signals

In our study of discrete-time signals and systems there are a number of basic signals that appear often and play an important role. These signals are defined below.

1. The *unit sample sequence* is denoted as $\delta(n)$ and is defined as

$$\delta(n) \equiv \begin{cases} 1, & \text{for } n = 0 \\ 0, & \text{for } n \neq 0 \end{cases} \quad (2.1.6)$$

In words, the unit sample sequence is a signal that is zero everywhere, except at $n = 0$ where its value is unity. This signal is sometimes referred to as a *unit impulse*. In contrast to the analog signal $\delta(t)$, which is also called a unit impulse and is defined to be zero everywhere except $t = 0$, and has unit area, the unit sample sequence is much less mathematically complicated. The graphical representation of $\delta(n)$ is shown in Fig. 2.2.

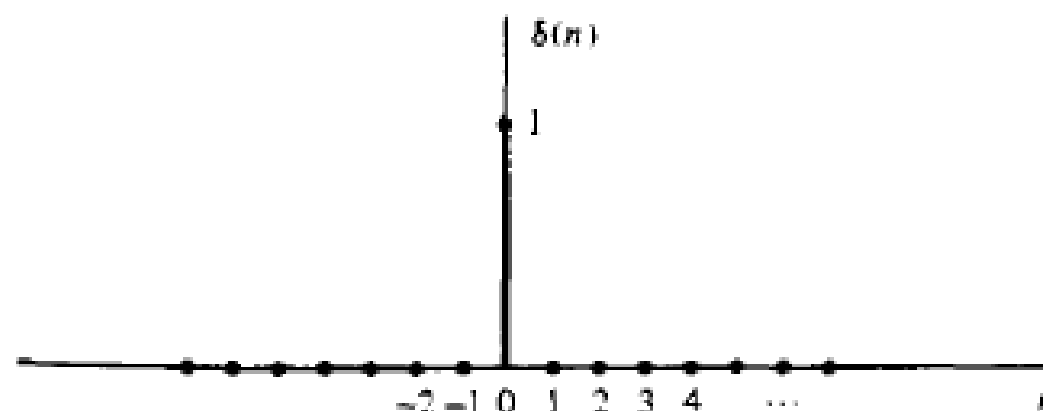


Figure 2.2 Graphical representation of the unit sample signal.

2. The *unit step signal* is denoted as $u(n)$ and is defined as

$$u(n) \equiv \begin{cases} 1, & \text{for } n \geq 0 \\ 0, & \text{for } n < 0 \end{cases} \quad (2.1.7)$$

Figure 2.3 illustrates the unit step signal.

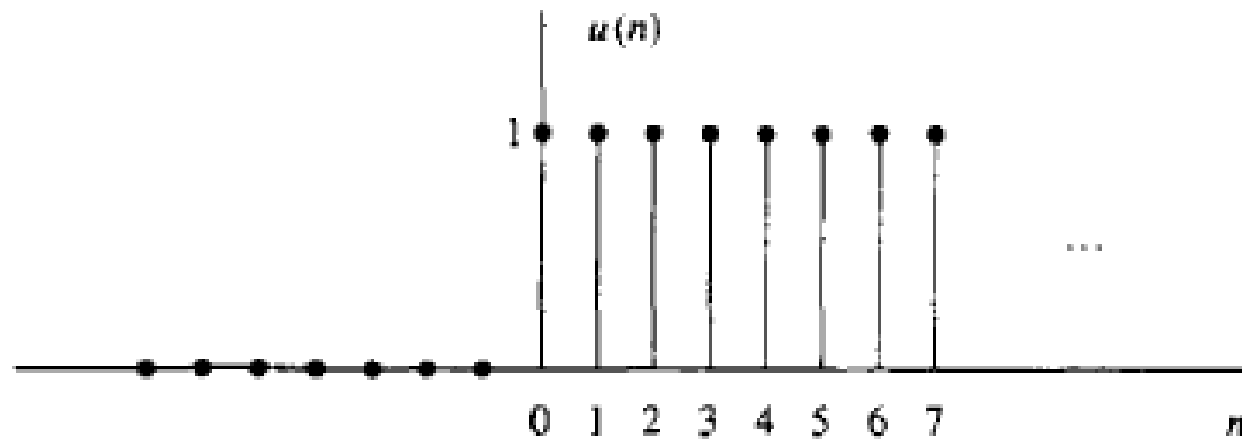


Figure 2.3 Graphical representation of the unit step signal.

3. The *unit ramp signal* is denoted as $u_r(n)$ and is defined as

$$u_r(n) \equiv \begin{cases} n, & \text{for } n \geq 0 \\ 0, & \text{for } n < 0 \end{cases} \quad (2.1.8)$$

This signal is illustrated in Fig. 2.4.

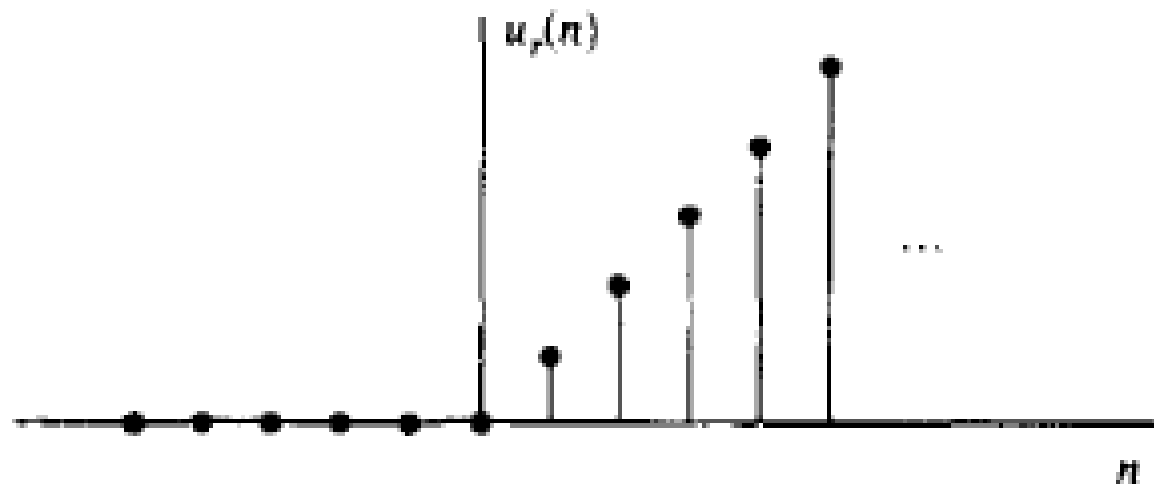


Figure 2.4 Graphical representation of the unit ramp signal.

4. The *exponential signal* is a sequence of the form

$$x(n) = a^n \quad \text{for all } n \quad (2.1.9)$$

If the parameter a is real, then $x(n)$ is a real signal. Figure 2.5 illustrates $x(n)$ for various values of the parameter a .

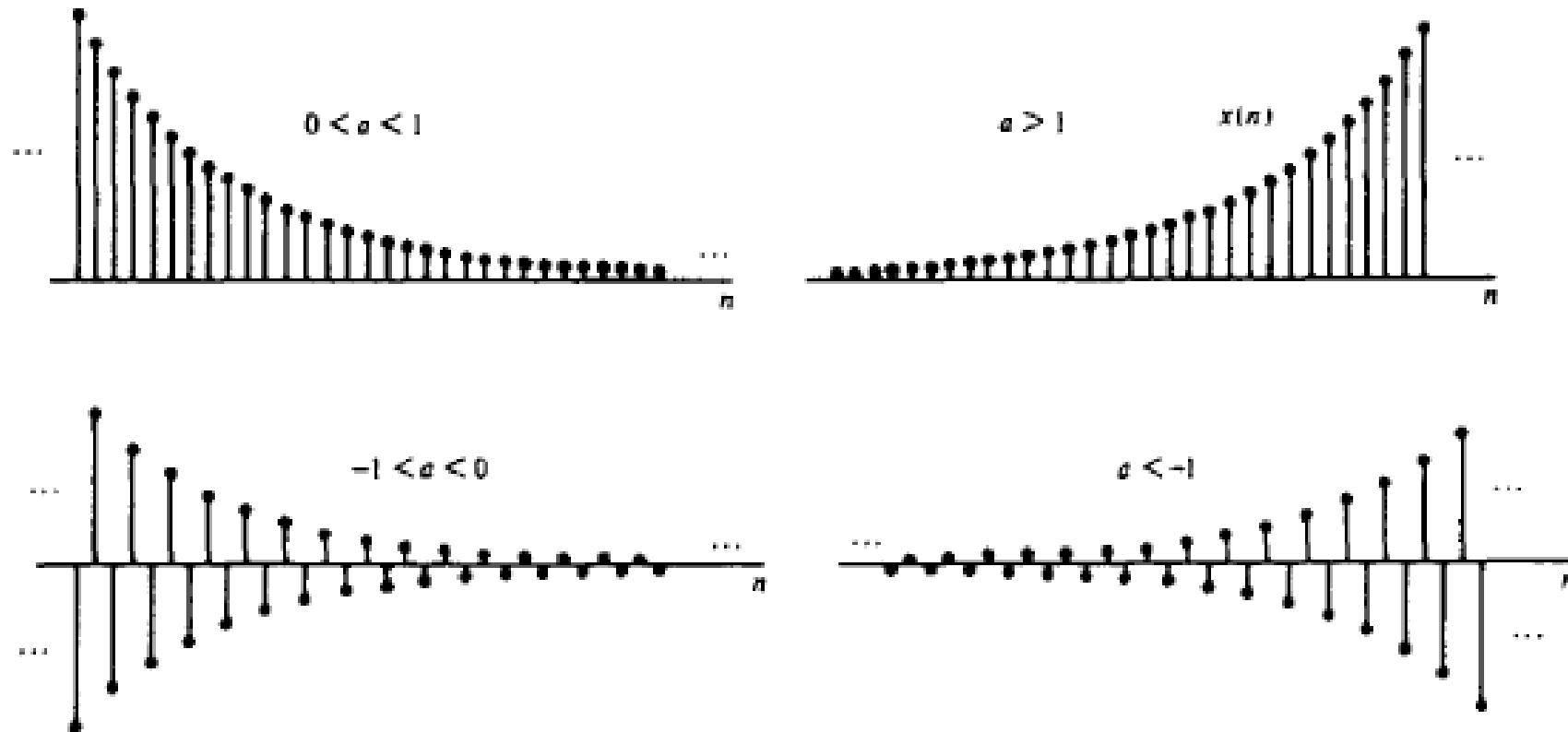
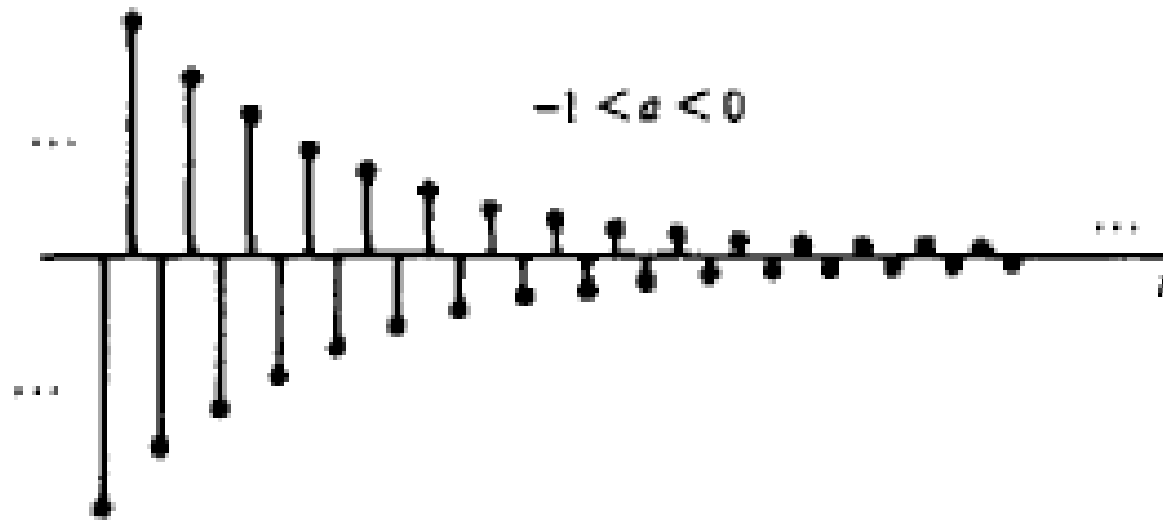


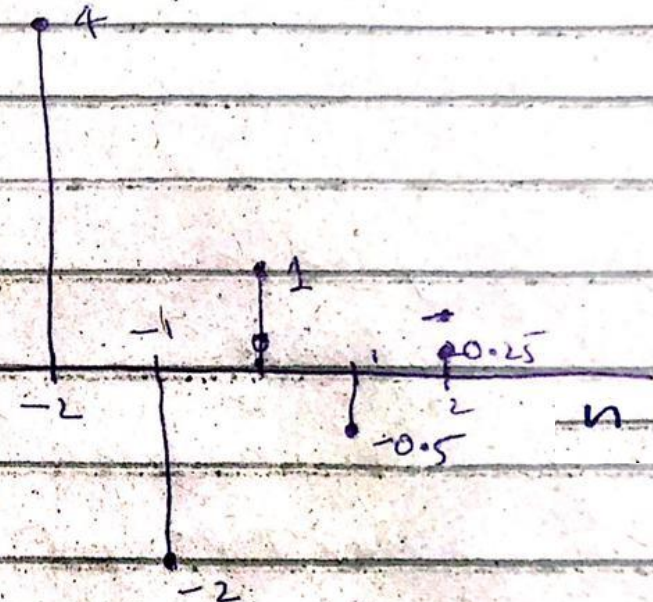
Figure 2.5 Graphical representation of exponential signals.

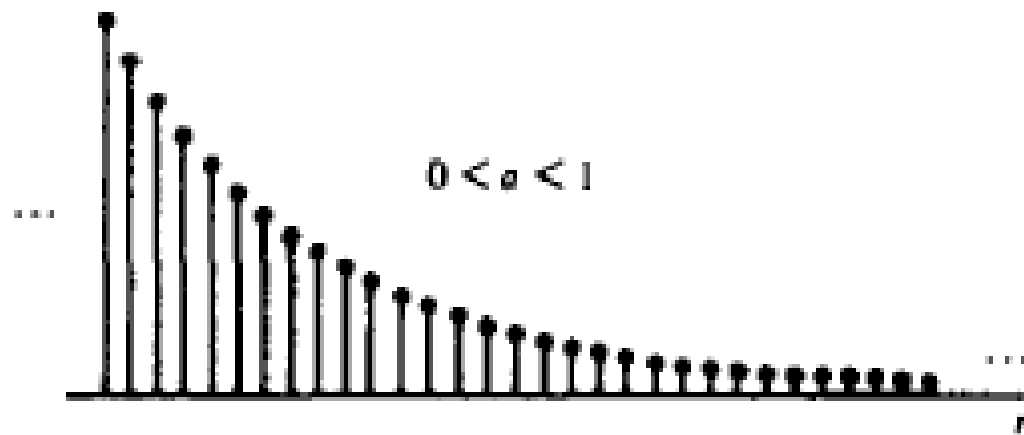
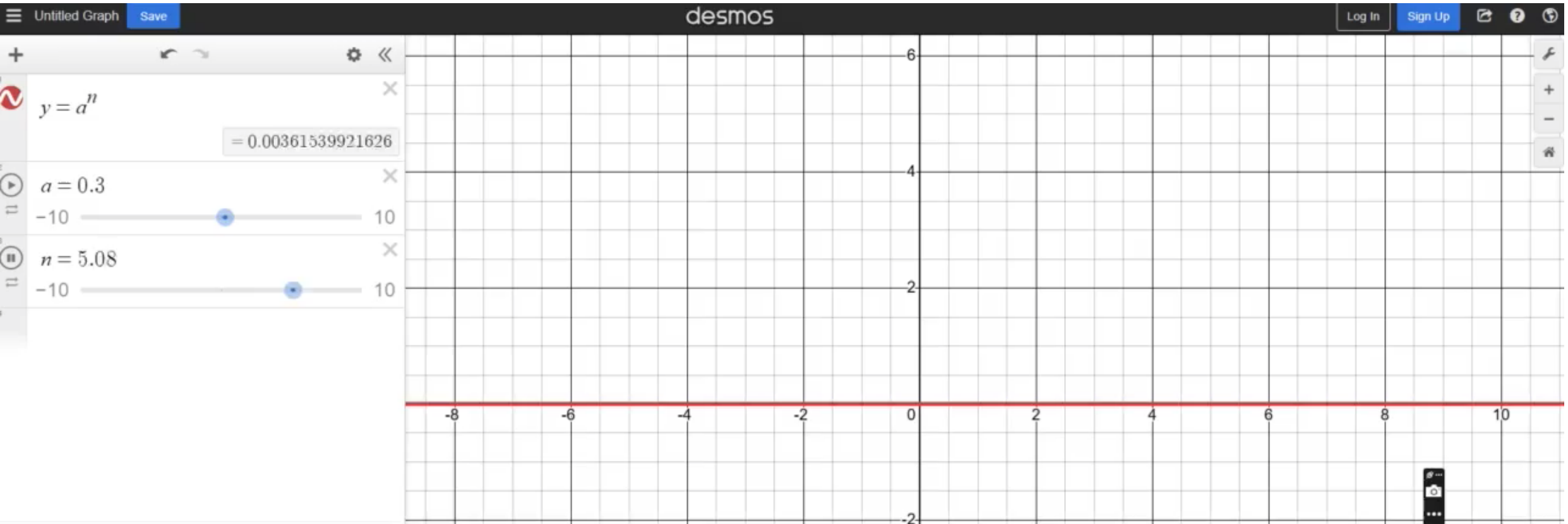


$x(n] = a^n$ for all n

Let $n = -2 : 1 : 2$
 $a = -0.5$

n	a^n
-2	4
-1	-2
0	1
1	-0.5
2	0.25





When the parameter a is complex valued, it can be expressed as

$$a \equiv r e^{j\theta}$$

where r and θ are now the parameters. Hence we can express $x(n)$ as

$$\begin{aligned} x(n) &= r^n e^{j\theta n} \\ &= r^n (\cos \theta n + j \sin \theta n) \end{aligned} \tag{2.1.10}$$

Since $x(n)$ is now complex valued, it can be represented graphically by plotting the real part

$$x_R(n) \equiv r^n \cos \theta n \tag{2.1.11}$$

as a function of n , and separately plotting the imaginary part

$$x_I(n) \equiv r^n \sin \theta n \tag{2.1.12}$$

as a function of n .

Figure 2.6 illustrates the graphs of $x_R(n)$ and $x_I(n)$ for $r = 0.9$ and $\theta = \pi/10$. We observe that the signals $x_R(n)$ and $x_I(n)$ are a damped (decaying exponential) cosine function and a damped sine function. The angle variable θ is simply the frequency of the sinusoid, previously denoted by the (normalized) frequency variable ω . Clearly, if $r = 1$, the damping disappears and $x_R(n)$, $x_I(n)$, and $x(n)$ have a fixed amplitude, which is unity.

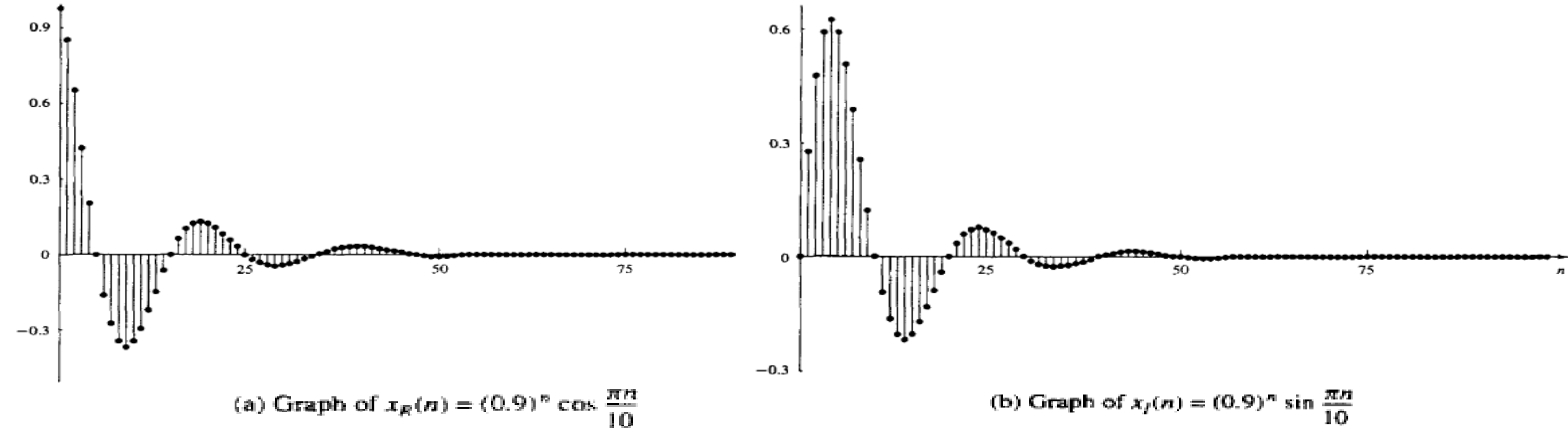


Figure 2.6 Graph of the real and imaginary components of a complex-valued exponential signal.

Alternatively, the signal $x(n)$ given by (2.1.10) can be represented graphically by the amplitude function

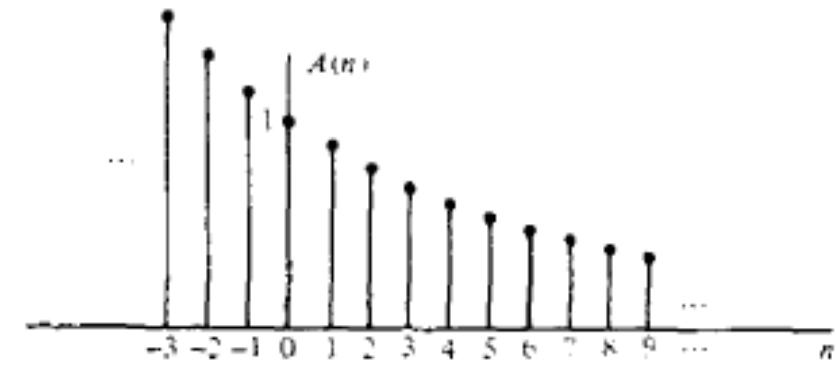
$$|x(n)| = A(n) \equiv r^n \quad (2.1.13)$$

and the phase function

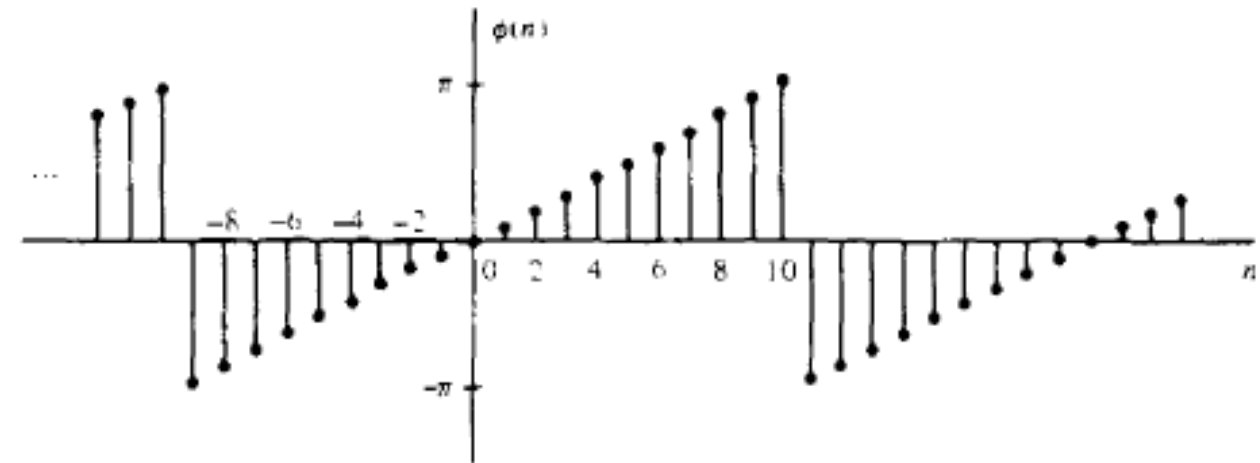
$$\angle x(n) = \phi(n) \equiv \theta n \quad (2.1.14)$$

Figure 2.7 illustrates $A(n)$ and $\phi(n)$ for $r = 0.9$ and $\theta = \pi/10$. We observe that the phase function is linear with n . However, the phase is defined only over the interval $-\pi < \theta \leq \pi$ or, equivalently, over the interval $0 \leq \theta < 2\pi$. Consequently, by convention $\phi(n)$ is plotted over the finite interval $-\pi < \theta \leq \pi$ or $0 \leq \theta < 2\pi$. In other words, we subtract multiples of 2π from $\phi(n)$ before plotting.

The subtraction of multiples of 2π from $\phi(n)$ is equivalent to interpreting the function $\phi(n)$ as $\phi(n)$, modulo 2π . The graph for $\phi(n)$, modulo 2π , is shown in Fig. 2.7b.



(a) Graph of $A(n) = r^n$, $r = 0.9$



(b) Graph of $\phi(n) = \frac{\pi}{10}n$, modulo 2π plotted in the range $(-\pi, \pi]$

Figure 2.7 Graph of amplitude and phase function of a complex-valued exponential signal: (a) graph of $A(n) = r^n$, $r = 0.9$; (b) graph of $\phi(n) = (\pi/10)n$, modulo 2π plotted in the range $(-\pi, \pi]$.

$\Rightarrow$ Phase is defined only over the interval $-\pi \leq \theta \leq \pi$

OR $0 \leq \theta \leq 2\pi$.

Without restricting the phase,

the phase $\phi = \theta n$ can grow

indefinitely as 'n' increases.

Therefore, we map the phase
to the range $[-\pi, \pi)$.

$$\Phi_{\text{normalized}} = \phi(n) - 2\pi \text{round}\left(\frac{\phi(n)}{2\pi}\right)$$

Suppose you have a phase signal $\phi(n)$
with values:

$$\phi(n) = \left\{ -3\pi, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{5\pi}{2}, 3\pi \right\}$$

without normalization, the values
span a wide range (-3π to 3π)

After normalization, the values
span a ~~wide~~ range between

$-\pi$ & π :

$$\left\{ \pi, -\frac{\pi}{2}, \frac{\pi}{2}, -\frac{\pi}{2}, \pi \right\}$$

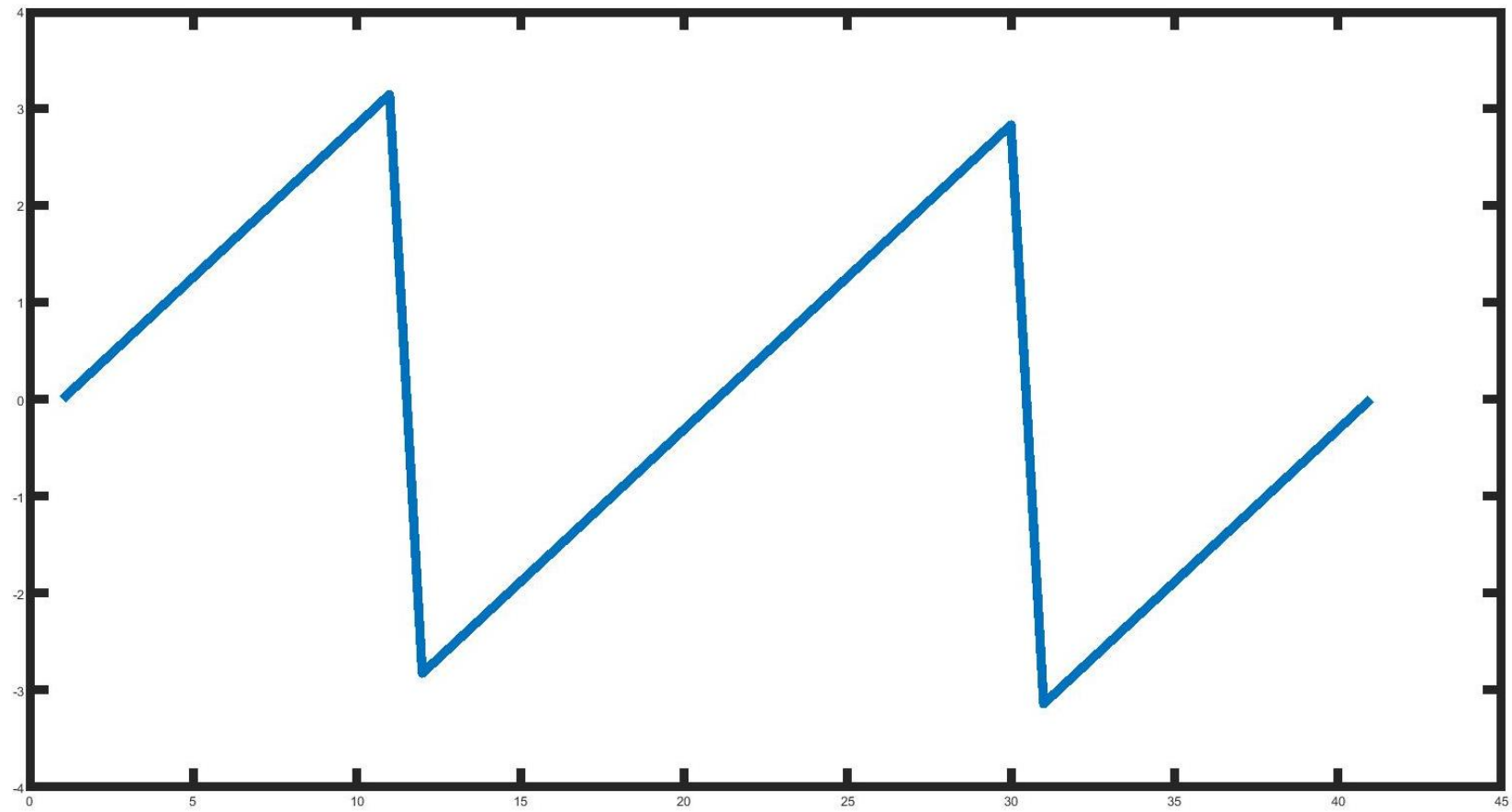
$$\begin{aligned} &= -3\pi - 2\pi \operatorname{round}\left(\frac{-3\pi}{2\pi}\right) \\ &= -3\pi - 2\pi(-1.5) \\ &= -3\pi + 4\pi = \pi \quad \checkmark \end{aligned}$$

$$\phi(n) = \frac{\pi}{10} n \text{ modulo } 2\pi$$

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Editor - Untitled*
Untitled* x +
1 clear
2 clc
3 n=-20:20;
4 y=(pi/10)*(n)-2*pi*round(n/20);
5 plot(y)

```



2.1.2 Classification of Discrete-Time Signals

Energy signals and power signals. The energy E of a signal $x(n)$ is defined as

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 \quad (2.1.15)$$

The energy of a signal can be finite or infinite. If E is finite (i.e., $0 < E < \infty$), then $x(n)$ is called an *energy signal*.

Many signals that possess infinite energy, have a finite average power. The average power of a discrete-time signal $x(n)$ is defined as

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 \quad (2.1.16)$$

2.1.2 Classification of Discrete-Time Signals

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 \quad (2.1.16)$$

If we define the signal energy of $x(n)$ over the finite interval $-N \leq n \leq N$ as

$$E_N \equiv \sum_{n=-N}^N |x(n)|^2 \quad (2.1.17)$$

then we can express the signal energy E as

$$E \equiv \lim_{N \rightarrow \infty} E_N \quad (2.1.18)$$

and the average power of the signal $x(n)$ as

$$P \equiv \lim_{N \rightarrow \infty} \frac{1}{2N+1} E_N \quad (2.1.19)$$

Clearly, if E is finite, $P = 0$. On the other hand, if E is infinite, the average power P may be either finite or infinite. If P is finite (and nonzero), the signal is called a *power signal*. The following example illustrates such a signal.

$$\underline{E = \text{Finite} ; P = 0}$$

Let

$$E = \sum_{n=-N}^N |x(n)|^2$$

Finite Energy

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$

$$P = \frac{1}{\infty} \approx 0$$

Example 2.1.1

Determine the power and energy of the unit step sequence. The average power of the unit step signal is

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N u^2(n)$$

Try it by yourself once!!!

Example 2.1.1

Determine the power and energy of the unit step sequence. The average power of the unit step signal is

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N u^2(n)$$
$$= \lim_{N \rightarrow \infty} \frac{N+1}{2N+1} = \lim_{N \rightarrow \infty} \frac{1 + 1/N}{2 + 1/N} = \frac{1}{2}$$

Is it correct? See next slide.

Consequently, the unit step sequence is a power signal. Its energy is infinite.

Let $N = 2$;

then

$$\sum_{n=0}^2 u^2(n)$$

$$u^2(0) + u^2(1) + u^2(2)$$

$$(1)^2 + (1)^2 + (1)^2$$

$$3 \rightarrow N+1$$

2. Energy of the Sequence

The energy of a discrete-time signal $x(n)$ is given by:

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

For the unit step sequence $u(n)$, we compute its energy E_u as follows:

$$E_u = \sum_{n=-\infty}^{\infty} |u(n)|^2$$

Since $u(n) = 0$ for $n < 0$ and $u(n) = 1$ for $n \geq 0$, we only need to sum over $n \geq 0$:

$$E_u = \sum_{n=0}^{\infty} |1|^2 = \sum_{n=0}^{\infty} 1$$

The sum of ones from $n = 0$ to infinity diverges, so:

$$E_u = \infty$$

Thus, the **energy** of the unit step sequence is infinite.

Consequently, the unit step sequence is a power signal. Its energy is infinite.

Similarly, it can be shown that the complex exponential sequence $x(n) = Ae^{j\omega_0 n}$ has average power A^2 , so it is a power signal. On the other hand, the unit ramp sequence is neither a power signal nor an energy signal.